

Trees Still Beat Transformers for Day-Ahead Electricity Price Forecasting

DATA

PJM + ERCOT
2019–2025 (60K hrs)
50 features

18 CONFIGURATIONS

Trees · DL · Transformers
Chronos-Bolt v1 & v2
+ Rolling Retraining

EVALUATION

Point · Probabilistic
Economic (PnL, Sharpe)
DM Tests + BH Correction

UNCERTAINTY

CQR · QRF · MC Dropout
Conformal fails under
regime shift (PICP 82.6%)

KEY FINDINGS

LightGBM Dominates

MAE: 3.05 (PJM) / 2.58 (ERCOT)
Beats all DL by 40–130%

Covariates > Architecture

Chronos-2 + covariates: 3.48
49% improvement over univar.

Rolling Retrain: +24%

Monthly retraining essential
4.03 → 3.05 MAE (PJM)

CQR Fails Under Shift

QRF: 91.2% PICP (on target)
CQR: 82.6% (misses by 7pp)

Rolling LightGBM achieves MAE = 3.05 \$/MWh (PJM) and 2.58 \$/MWh (ERCOT)

outperforming 17 alternatives including Chronos-2, PatchTST, iTransformer, and TFT across 2 US markets (2024–2025)